Global Vulnerability Database for Improved Regional Risk Assessments

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Abstract

This study presents a comprehensive global vulnerability database aimed at improving regional seismic risk assessments through tackling the main limitations of existing databases in the literature, which typically rely on overly simplified models that do not capture storey-level seismic demands and certain response patterns. The database encompasses over 1,400 building classes and leverages a refined numerical modeling approach that uses “stick-and-mass” multi-degree-of-freedom oscillators. Its key contributions include: (1) an extensive suite of updated vulnerability functions for assessing structural damage across various building types; (2) region-specific nonstructural vulnerability functions derived based on regional storey-loss functions that utilize storey-level demands, such as drifts and floor accelerations, to deliver more robust loss estimates due to non-structural damage; (3) a set of vulnerability functions specifically addressing losses in building contents. To showcase the utility of this new database, we use its outputs to conduct global risk analyses and compute key risk metrics essential for effective earthquake risk mitigation strategies.